GlucoTwin AI 2.0

Personalized Diabetes Digital Twin · Happiest Health Digital Twin Challenge 2026

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- Research prototype · all included demo data is synthetic
- Not a medical device; not for diagnosis or care decisions

The healthcare problem

Glucose varies over time; retrospective review may not express a near-term individualized pattern.

- Historical health context and current readings are often reviewed separately.
- A useful prototype should make time, data freshness, and missingness visible.
- This project demonstrates a technical workflow, not proven patient benefit.

Proposed solution

Link an anonymous profile to historical context and a dynamic observation timeline.

- Store age, BMI, diabetes duration, and HbA1c only when available.
- Append timestamped glucose, heart rate, sleep, and step counts.
- Estimate a bounded future event and return input timestamp, warnings, and a safety disclaimer.

Why a digital twin?

An inspectable, updateable representation of a person's stored state.

- Historical attributes and changing observations remain linked by anonymous patient ID.
- Clinicians/evaluators can inspect the records behind a model output.
- This is a data twin proof of concept, not a full physiological replica.

Two data streams

Static context + dynamic time-series observations.

- Static: age, BMI, optional HbA1c and diabetes duration.
- Dynamic: timezone-aware glucose and optional heart rate, sleep, and activity.
- Synthetic trajectories are generated reproducibly and explicitly marked; no public or patient data is bundled.

System architecture

Validate → persist → engineer past-only features → infer → inspect.

- FastAPI request schemas validate units, ranges, timestamps, and duplicate source records.
- SQLite stores anonymous profiles, readings, and predictions.
- A separate offline pipeline saves a trusted joblib artifact and held-out evaluation report.
- See docs/architecture_diagram.pdf and editable docs/architecture_diagram.svg.

Dataset and ethics

48,384 synthetic readings · 36 generated profiles · 14 days · 15-minute intervals.

- Fixed seed 20260626; simplified meal pulses, circadian variation, noise, activity, sleep, and static attributes.
- Assumptions are software-test fixtures, not a clinical simulator.
- No external/public dataset was supplied or used; all demo records are labelled synthetic.

Past-only feature engineering

Use current and earlier observations as inputs; reserve future glucose for labels only.

- Current glucose, 15- and 60-minute changes, 60-minute mean, and 120-minute variability.
- Heart rate, sleep, trailing steps, optional BMI/HbA1c, and time-of-day encoding.
- Training-fitted median imputation; irregular 15-minute training cadence is rejected.

Target and training design

Prototype label: any of the next eight 15-minute glucose readings reaches ≥180 mg/dL.

- This is an explicit project target, not a universal medical standard.
- Patient-group holdout: 21 train, 7 validation, 8 test profiles.
- Prior baseline, Logistic Regression, and Random Forest compared; selected on validation PR-AUC.
- Training-only class weighting; no test-set tuning; default threshold 0.5 is reported.

Model selection and results

Actual generated results are reported only when the evaluation artifact exists.

- Validation PR-AUC: prior 0.2630; logistic 0.6577; Random Forest 0.8548.
- Selected by validation PR-AUC: random_forest.
- Held-out test at fixed 0.5: precision 0.7781; recall 0.5590; F1 0.6506.
- PR-AUC 0.7983; ROC-AUC 0.9339; n=10,688 rows from 8 held-out synthetic profiles.
- Confusion matrix [[TN, FP], [FN, TP]]: [[8106, 355], [982, 1245]].
- These are synthetic-only results, not clinical validation; the test was not used for model selection.

Prediction workflow

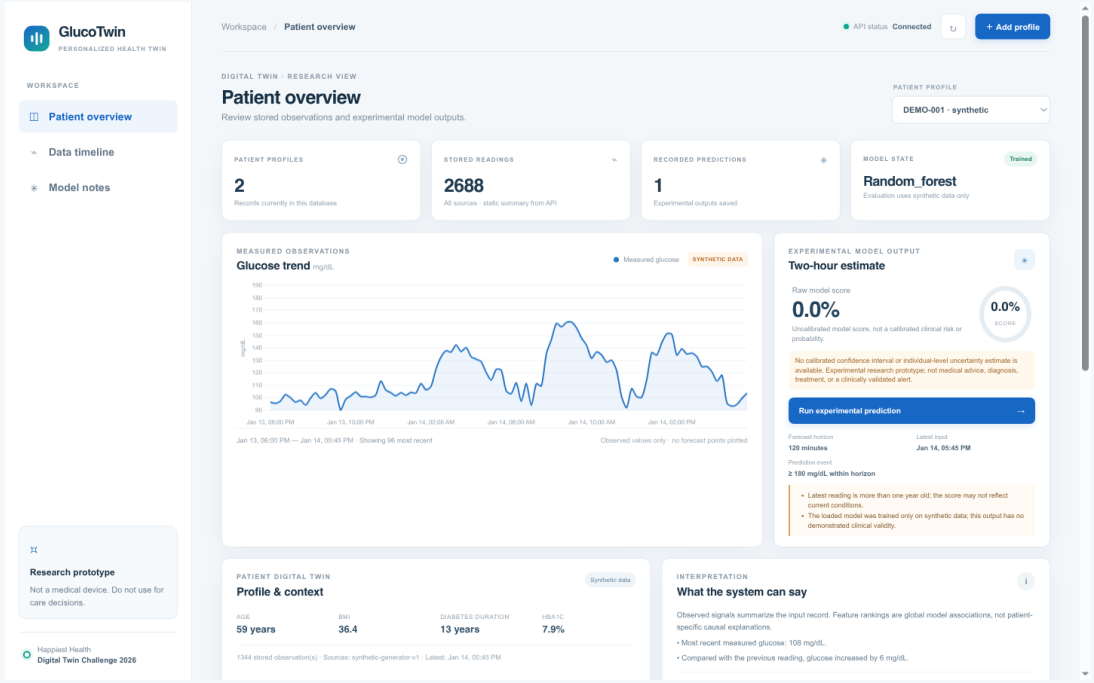
Latest readings → past-only feature row → model score → persisted response.

- API returns patient ID, prediction time, 120-minute horizon, event definition, and model version.
- Response includes uncalibrated raw score, input timestamp, observed signals, global model ranking, quality warnings, and disclaimer.
- If training was not completed, inference returns a clear service error and training command.

Dashboard walkthrough

The browser dashboard reads patient, observations, score history, and evaluation from FastAPI.

- Select backend-returned profiles; inspect measured-only glucose chart and source timeline.
- Add a manual observation without a fake sensor connection.
- Review stale/missing context, synthetic labels, score limitations, and backend evaluation.



Synthetic patient example

A generated synthetic profile; the prediction is an experimental model output.

- Generated synthetic profile SYN-033; measured input at 2025-01-10T13:00:00Z was 176.5 mg/dL.
- Actual saved model returned raw score 0.9916; uncalibrated, not an individual clinical probability.
- The simulated next-eight-row target is positive at this point. This historical synthetic example is stale and is not a current reading or care recommendation.

Limitations and next steps

The current score does not establish safety, usefulness, or generalization.

- Only synthetic data; simplified generator; one project-defined threshold; small number of held-out synthetic profiles.
- No external validation, calibration, confidence interval, subgroup analysis, clinical review, or prospective evaluation.
- No authentication or production security; SQLite is local demonstration storage.
- Next: licensed representative data, ethics/clinical review, external validation, calibration, subgroup evaluation, and security controls.

Conclusion and references

A reproducible full-stack technical proof of concept, not a clinical product.

- Project documentation: README, model card, data dictionary, limitations, and 21-minute demo script.
- References: FastAPI documentation; scikit-learn documentation; SQLite documentation; Chart.js documentation.
- Team leader must verify submission rules, add only confirmed team details, and insert actual repository/video links.
- Thank you · Happiest Health Digital Twin Challenge 2026